# MATHEMATICAL METHODS IN BIOENGINEERING
# EXAM # 1

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**P1.** In a bioreactor engineering experiment, the rate of biomass accumulation of a bacterial culture over the first 4 hours is modeled by the function:

$$r(t) = 3t^2 + 2t$$

where $t$ is the time in hours ($h$) and $r(t)$ is the growth rate in grams per hour ($g/h$). The total mass of the population accumulated between $t = 0$ and $t = 2$ hours is given by the definite integral:

$$M = \int_0^2 (3t^2 + 2t)\ \mathrm{d}t$$

Calculate the total biomass $M$ (in grams) accumulated at $t = 2$ hours.

Sol: To find the total biomass accumulated, we evaluate the definite integral by finding the antiderivative:

$$M = \int_0^2 (3t^2 + 2t)\ \mathrm{d}t = \left[t^3 + t^2\right]_0^2$$

Substituting the upper and lower limits:

$$M = (2^3 + 2^2) - (0^3 + 0^2) = (8 + 4) - 0 = 12 \text{ grams}$$

Thus, the total accumulated biomass at $t = 2$ hours is 12 grams.

**P2.** During a drug delivery assay, the concentration $C(t)$ of a therapeutic agent in a patient's bloodstream (measured in mg/L) as a function of time $t$ (in hours) is modeled by the rational function:

$$C(t) = \frac{8t^2 + 5t}{2t^2 + 3}$$

Biomedical engineers are interested in finding the steady-state concentration, which corresponds to the limiting value of the concentration as time approaches infinity. Evaluate the limit:

$$L = \lim_{t\to\infty} \frac{8t^2 + 5t}{2t^2 + 3}$$

Sol: To evaluate the limit as $t \to \infty$, we divide both the numerator and the denominator by the highest power of $t$ in the denominator, which is $t^2$:

$$L = \lim_{t\to\infty} \frac{\frac{8t^2}{t^2} + \frac{5t}{t^2}}{\frac{2t^2}{t^2} + \frac{3}{t^2}} = \lim_{t\to\infty} \frac{8 + \frac{5}{t}}{2 + \frac{3}{t^2}}$$

As $t \to \infty$, the terms $\frac{5}{t} \to 0$ and $\frac{3}{t^2} \to 0$. Therefore:

$$L = \frac{8 + 0}{2 + 0} = 4 \text{ mg/L}$$

The steady-state concentration of the therapeutic agent is 4 mg/L.

**P3.** (a) In a tissue engineering laboratory, electrospun polymer nanofibers are fabricated to serve as scaffolds for cell growth. The diameter of these fibers, denoted by $Y$, is known to be normally distributed with a mean of $\mu = 500$ nm and a standard deviation of $\sigma = 50$ nm (i.e., $Y \sim N(500, 50)$). Biomedical studies indicate that optimal cell attachment and proliferation occur only on fibers with diameters between 440 nm and 610 nm. Estimate the percentage of the fabricated nanofiber population that satisfies the optimal structural criteria for cell growth.

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Sol: We want to find the probability $P(440 \leq Y \leq 610)$. First, we standardize the variables to the standard normal distribution $Z \sim N(0,1)$ using $Z = \frac{Y-\mu}{\sigma}$:

$$Z_1 = \frac{440 - 500}{50} = \frac{-60}{50} = -1.2$$

$$Z_2 = \frac{610 - 500}{50} = \frac{110}{50} = 2.2$$

Thus, the probability is:

$$P(-1.2 \leq Z \leq 2.2) = \Phi(2.2) - \Phi(-1.2)$$

Using standard normal distribution tables:

$$\Phi(2.2) \approx 0.9861, \quad \Phi(-1.2) \approx 0.1151$$

$$P(-1.2 \leq Z \leq 2.2) = 0.9861 - 0.1151 = 0.8710$$

Therefore, approximately 87.10% of the nanofiber population meets the optimal structural criteria.

(b) Let $X$ be a normally distributed random variable, $X \sim N(\mu, \sigma)$ for some real number $\mu$ and some $\sigma > 0$. Define

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left( - \frac{(x-\mu)^2}{2\sigma^2} \right).$$

In this problem you can use the following (you may assume that these properties are already known):

- For any real number $a$,
$$P(X < a) = \int_{-\infty}^{a} f(x)\, \mathrm{d}x$$
- $\displaystyle\int_{-\infty}^{\infty} z \exp(-z^2/2)\, \mathrm{d}z = 0.$
- $\displaystyle\int_{-\infty}^{\infty} (x-\mu)\, f(x)\, \mathrm{d}x = \int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x - \mu \int_{-\infty}^{\infty} f(x)\, \mathrm{d}x.$
- The quantity $\displaystyle\int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x$ is known as *the expected value of the random variable X.*

Now you are asked to answer/do the following:

(i) What is the value of $\displaystyle\int_{-\infty}^{\infty} f(x)\, \mathrm{d}x$? (An heuristic justification is enough.)

(ii) Using a change of variable, prove that:
$$\int_{-\infty}^{\infty} (x-\mu) f(x)\, \mathrm{d}x = \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z \exp(-z^2/2)\, \mathrm{d}z.$$

(iii) Show how to combine the above properties (even if you were not able to prove them), to establish that

*The expected value of X is $\mu$.*

Sol:

(a) The value of $\displaystyle\int_{-\infty}^{\infty} f(x)\, \mathrm{d}x = 1$. Heuristically, $f(x)$ is the probability density function (PDF) of a continuous random variable $X$. Since the total probability over the entire real line must equal 100%, the total area under the density curve is exactly 1.

(b) Let us apply the substitution $z = \frac{x-\mu}{\sigma}$. This implies $x - \mu = \sigma z$ and $\mathrm{d}x = \sigma\, \mathrm{d}z$. The limits of integration remain $-\infty$ and $\infty$. Substituting these into the integral gives:
$$\int_{-\infty}^{\infty} (x-\mu) f(x)\, \mathrm{d}x = \int_{-\infty}^{\infty} (\sigma z) \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{z^2}{2}\right) \sigma\, \mathrm{d}z$$
Canceling out $\sigma$ inside the fraction and pulling out constants:
$$= \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z \exp(-z^2/2)\, \mathrm{d}z$$

(c) From property (3) and the definitions given:

$$\int_{-\infty}^{\infty} (x-\mu)\, f(x)\, \mathrm{d}x = \text{Expected Value} - \mu \int_{-\infty}^{\infty} f(x)\, \mathrm{d}x$$

Using the result from (b) and property (2), the left-hand side simplifies to:

$$\frac{\sigma}{\sqrt{2\pi}} \cdot (0) = 0$$

Using the result from (a), the second term on the right-hand side simplifies because $\int_{-\infty}^{\infty} f(x)\, \mathrm{d}x = 1$. Thus, the equation becomes:

$$0 = \text{Expected Value} - \mu \cdot (1) \implies \text{Expected Value} = \mu$$

**P4.** The logistic growth of a microbial population is described by the differential equation $\frac{\mathrm{d}N}{\mathrm{d}t} = rN\left(1 - \frac{N}{K}\right)$. Given the initial condition $N(0) = N_0$, solving this equation for the intrinsic growth rate $r$ yields the following algebraic relationship:

$$r = \frac{1}{t}\ln\left(\frac{K - N_0}{N_0}\right) + \frac{1}{t}\ln\left(\frac{N(t)}{K - N_t}\right)$$

Suppose a bioengineer monitors a cell culture population and determines that the carrying capacity is $K = 30$ million cells. The population data at key checkpoints is recorded as follows:

$$N(0) = 10 \text{ million cells} \quad \text{and} \quad N(5) = 22 \text{ million cells.}$$

(a) Use the provided formula and the data points at $t = 0$ and $t = 5$ to estimate the exact value of the intrinsic growth rate $r$. *(Leave your answer in terms of natural logarithms or round to four decimal places.)*
(b) Using your calculated value of $r$ and the explicit solution model, project the expected cell population size at $t = 10$ hours.

Sol:
(a) Substituting $t = 5$, $K = 30$, $N_0 = 10$, and $N(5) = 22$ into the formula:

$$r = \frac{1}{5}\ln\left(\frac{30 - 10}{10}\right) + \frac{1}{5}\ln\left(\frac{22}{30 - 22}\right)$$

$$r = \frac{1}{5}\ln(2) + \frac{1}{5}\ln\left(\frac{22}{8}\right) = \frac{1}{5}\left(\ln(2) + \ln(2.75)\right) = \frac{1}{5}\ln(5.5)$$

Thus, exactly $r = \frac{\ln(5.5)}{5} \approx 0.3409 \text{ h}^{-1}$.
(b) Using the same formula for $t = 10$, we can solve for $N(10)$:

$$r = \frac{1}{10}\ln(2) + \frac{1}{10}\ln\left(\frac{N(10)}{30 - N(10)}\right)$$

Multiply by 10:

$$10r = \ln(2) + \ln\left(\frac{N(10)}{30 - N(10)}\right)$$

Since $10r = 10\left(\frac{\ln(5.5)}{5}\right) = 2\ln(5.5) = \ln(5.5^2) = \ln(30.25)$, we substitute this back:

$$\ln(30.25) = \ln\left(\frac{2N(10)}{30 - N(10)}\right) \implies 30.25 = \frac{2N(10)}{30 - N(10)}$$

$$30.25(30 - N(10)) = 2N(10) \implies 907.5 - 30.25N(10) = 2N(10)$$

$$907.5 = 32.25N(10) \implies N(10) = \frac{907.5}{32.25} \approx 28.14 \text{ million cells}$$

**P5.** The current $I$ (in amperes, A) flowing through an AC circuit as a function of time $t$ (in seconds) is given by:

$$I = 120\sin(30\pi t), \quad t \geq 0$$

(a) What is the amplitude of the current function?
(b) What is the period of the current function?
(c) Sketch the graph of this function over a time interval covering exactly two full periods.

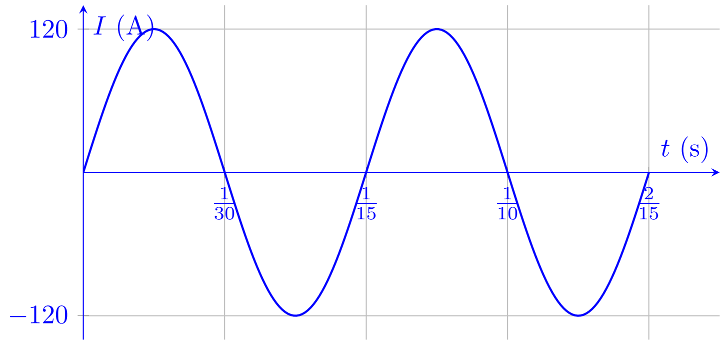
Sol:

(a) The amplitude is the maximum absolute value of the function, which is given by the coefficient of the sine term: 120 A.

(b) The angular frequency is $\omega = 30\pi$. The period $T$ is given by:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{30\pi} = \frac{1}{15} \text{ seconds} \approx 0.0667 \text{ s}$$

(c) Two full periods correspond to a total time interval of $2T = \frac{2}{15}$ seconds. Below is the representative sketch:




**P6.** (a) In a biomechanics study, engineers measure the relationship between the applied tensile force $x$ (in Newtons, N) and the resulting elongation $Y$ (in millimeters, mm) of a novel biocompatible hydrogel sample. The data collected from 4 trials is summari zed in the table below:

| **Force** $x_i$ (N) | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **Elongation** $Y_i$ (mm) | 2 | 3 | 5 | 6 |

We want to fit a linear regression line using the least-squares estimators $\hat{\beta}$ and $\hat{\alpha}$:

$$\hat{\beta} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \quad \hat{\alpha} = \bar{Y} - \hat{\beta}\bar{x}$$

where $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$ and $\bar{Y} = \frac{\sum_{i=1}^{n} Y_i}{n}$.

(i) Compute the sample means $\bar{x}$ and $\bar{Y}$.

(ii) Determine the exact values of the slope estimator $\hat{\beta}$ and the intercept estimator $\hat{\alpha}$.

(iii) Using the model $\hat{Y} = \hat{\alpha} + \hat{\beta}x$, predict the expected elongation $\hat{Y}$ of the hydrogel sample if a tensile force of $x = 2.5$ N is applied.

Sol:

(a) Given $n = 4$:

$$\bar{x} = \frac{1+2+3+4}{4} = \frac{10}{4} = \mathbf{2.5}$$

$$\bar{Y} = \frac{2+3+5+6}{4} = \frac{16}{4} = \mathbf{4.0}$$

(b) Let's calculate the components for $\hat{\beta}$:

| $x_i$ | $Y_i$ | $x_i - \bar{x}$ | $Y_i - \bar{Y}$ | $(x_i - \bar{x})(Y_i - \bar{Y})$ | $(x_i - \bar{x})^2$ |
|---|---|---|---|---|---|
| 1 | 2 | -1.5 | -2.0 | 3.0 | 2.25 |
| 2 | 3 | -0.5 | -1.0 | 0.5 | 0.25 |
| 3 | 5 | 0.5 | 1.0 | 0.5 | 0.25 |
| 4 | 6 | 1.5 | 2.0 | 3.0 | 2.25 |
| $\sum$ | | | | 7.0 | 5.0 |

$$\hat{\beta} = \frac{7.0}{5.0} = 1.4$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{x} = 4.0 - (1.4)(2.5) = 4.0 - 3.5 = 0.5$$

(c) Using the prediction model $\hat{Y} = 0.5 + 1.4x$, for $x = 2.5$ N:

$$\hat{Y} = 0.5 + 1.4(2.5) = 0.5 + 3.5 = 4.0 \text{ mm}$$

(b) Let $S_{XY}$ be the quantity defined in the least-squares estimation formula:

$$S_{XY} = \sum_{i=1}^{n}(x_i - \bar{x})(Y_i - \bar{Y})$$

Mathematically prove the identity for the numerator shortcut formula:

$$S_{XY} = \sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y}$$

*(Hint: Expand the binomial expression inside the summation and distribute $\sum$. Recall that $\sum_{i=1}^{n} x_i = n\bar{x}$ and $\sum_{i=1}^{n} \bar{Y} = n\bar{Y}$.)*

Sol: Expanding the algebraic expression inside the summation gives:

$$S_{XY} = \sum_{i=1}^{n}(x_i Y_i - x_i\bar{Y} - \bar{x}Y_i + \bar{x}\bar{Y})$$

Distributing the summation operator $\sum$ across each term:

$$S_{XY} = \sum_{i=1}^{n} x_i Y_i - \sum_{i=1}^{n} x_i\bar{Y} - \sum_{i=1}^{n} \bar{x}Y_i + \sum_{i=1}^{n} \bar{x}\bar{Y}$$

Since $\bar{x}$ and $\bar{Y}$ are constants relative to the summation index $i$, they can be factored out:

$$S_{XY} = \sum_{i=1}^{n} x_i Y_i - \bar{Y}\sum_{i=1}^{n} x_i - \bar{x}\sum_{i=1}^{n} Y_i + n\bar{x}\bar{Y}$$

Using the hint properties $\sum_{i=1}^{n} x_i = n\bar{x}$ and $\sum_{i=1}^{n} Y_i = n\bar{Y}$, we substitute these values:

$$S_{XY} = \sum_{i=1}^{n} x_i Y_i - \bar{Y}(n\bar{x}) - \bar{x}(n\bar{Y}) + n\bar{x}\bar{Y}$$

$$S_{XY} = \sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y} - n\bar{x}\bar{Y} + n\bar{x}\bar{Y}$$

Simplifying the final constant terms yields the desired identity:

$$S_{XY} = \sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y}$$